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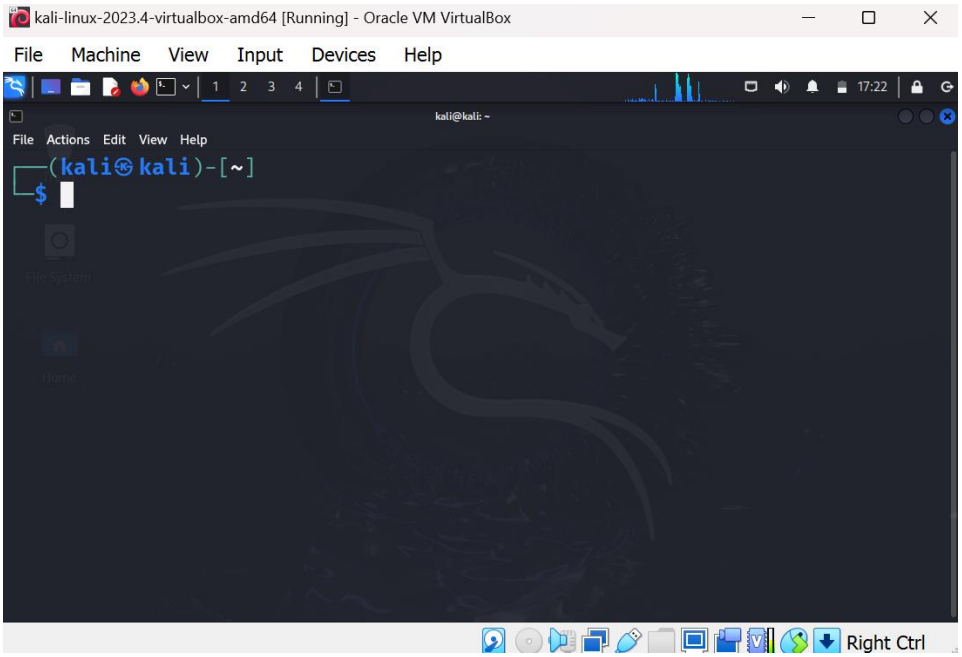
Manipulating files and directories in Linux

Lab 04

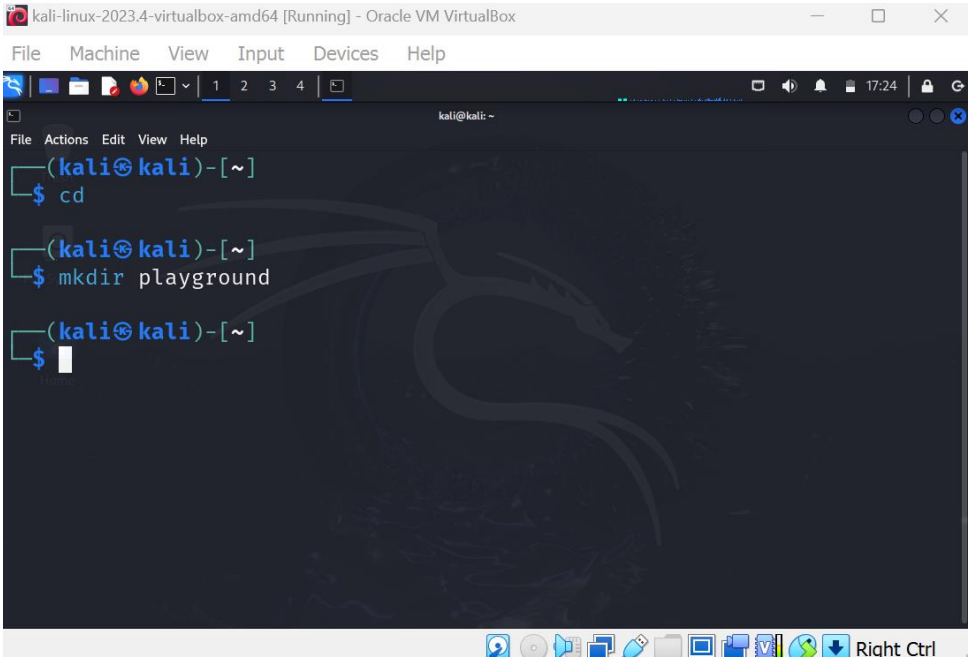
Objective: To manipulate files and directories in linux.

Introduction: In this lab, I will let learn to use the proper Linux commands to navigate between files and directories.

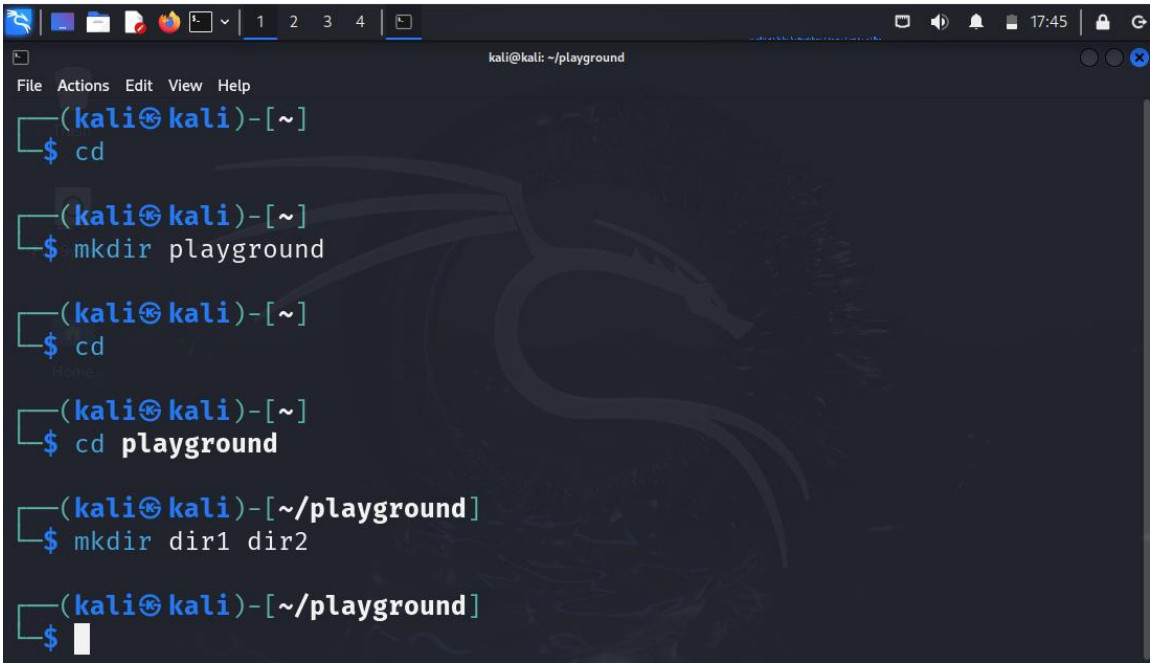
Step 1: After using the command, ctrl+alt+t, a nre terminal opened.



Step 2: After using the mkdir command, a new directory was made for playground.



Step 3: In the playground directory, I have created others directories.



Step 4: The file has been copied.

```
kali@kali: ~/playground
File  Actions  Edit  View  Help
└─$ cd playground

┌──(kali㉿kali)-[~/playground]
└─$ mkdir dir1 dir2

┌──(kali㉿kali)-[~/playground]
└─$ cp /etc/passwd .

┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 12
drwxr-xr-x 2 kali kali 4096 Apr 18 17:45 dir1
drwxr-xr-x 2 kali kali 4096 Apr 18 17:45 dir2
-rw-r--r-- 1 kali kali 3217 Apr 18 17:46 passwd

┌──(kali㉿kali)-[~/playground]
└─$
```

Step 5: The files are copied.

```
kali@kali: ~/playground
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 12
drwxr-xr-x 2 kali kali 4096 Apr 18 17:45 dir1
drwxr-xr-x 2 kali kali 4096 Apr 18 17:45 dir2
-rw-r--r-- 1 kali kali 3217 Apr 18 17:46 passwd

┌──(kali㉿kali)-[~/playground]
└─$ cp -v /etc/passwd .
'/etc/passwd' → './passwd'

┌──(kali㉿kali)-[~/playground]
└─$ cp -i /etc/passwd .
cp: overwrite './passwd'?

┌──(kali㉿kali)-[~/playground]
└─$
```

Step 6: In this step, I moved fub and rename it.

```
┌──(kali㉿kali)-[~/playground]
└─$ mv passwd fub

┌──(kali㉿kali)-[~/playground]
└─$ mv fun dir1
mv: cannot stat 'fun': No such file or directory

┌──(kali㉿kali)-[~/playground]
└─$ mv fub dir1

┌──(kali㉿kali)-[~/playground]
└─$ mv dir1/fub dir2

┌──(kali㉿kali)-[~/playground]
└─$ mv dir2/fub .

┌──(kali㉿kali)-[~/playground]
└─$ mv fub dir1
```

Step 7: I renamed and moved files. For example. I moved dir1 into dir2.

```
┌──(kali㉿kali)-[~/playground]
└─$ mv dir1 dir2

┌──(kali㉿kali)-[~/playground]
└─$ ls -l dir2
total 4
drwxr-xr-x 2 kali kali 4096 Apr 18 17:58 dir1

┌──(kali㉿kali)-[~/playground]
└─$ ls -l dir2/dir1
total 4
-rw-r--r-- 1 kali kali 3217 Apr 18 17:49 fub

┌──(kali㉿kali)-[~/playground]
└─$ mv dir2/dir1 .

┌──(kali㉿kali)-[~/playground]
└─$ mv dir1/fub .
```

Step 8: I created a hard link with dir1 and dir2 in the playground directory.

```
┌──(kali㉿kali)-[~/playground]
└─$ ln fun fun-hard

┌──(kali㉿kali)-[~/playground]
└─$ ln fun dir1/fun-hard

┌──(kali㉿kali)-[~/playground]
└─$ ln fun dir2/fun-hard

┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 16
drwxr-xr-x 2 kali kali 4096 Apr 24 19:19 dir1
drwxr-xr-x 2 kali kali 4096 Apr 24 19:19 dir2
-rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun
-rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun-hard

┌──(kali㉿kali)-[~/playground]
└─$ ls -li
total 16
1310724 drwxr-xr-x 2 kali kali 4096 Apr 24 19:19 dir1
1310725 drwxr-xr-x 2 kali kali 4096 Apr 24 19:19 dir2
1310726 -rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun
1310726 -rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun-hard

┌──(kali㉿kali)-[~/playground]
└─$
```

Step 9: In this step, I created a symbolic link between the given files.

```
┌──(kali㉿kali)-[~/playground]
└─$ ln -s fun fun-sym

┌──(kali㉿kali)-[~/playground]
└─$ ln -s ../fun dir1/fun-sym

┌──(kali㉿kali)-[~/playground]
└─$ ln -s ../fun dir2/fun-sym

┌──(kali㉿kali)-[~/playground]
└─$ ls -l dir1
total 4
-rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun-hard
lrwxrwxrwx 1 kali kali    6 Apr 24 19:24 fun-sym → ../fun

┌──(kali㉿kali)-[~/playground]
└─$ ln -s home/me/playground/fun dir1/fun-sym
ln: failed to create symbolic link 'dir1/fun-sym': File exists
```

Step 10: Here we are better able to view what happened.

```
┌──(kali㉿kali)-[~/playground]
└─$ ln -s dir1 dir1-sym

┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 16
drwxr-xr-x 2 kali kali 4096 Apr 24 19:24 dir1
lrwxrwxrwx 1 kali kali    4 Apr 24 19:32 dir1-sym → dir1
drwxr-xr-x 2 kali kali 4096 Apr 24 19:28 dir2
-rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun
-rw-r--r-- 4 kali kali 3217 Apr 18 17:49 fun-hard
lrwxrwxrwx 1 kali kali    3 Apr 24 19:24 fun-sym → fun
```

Step 11: Fun-hard has been removed.

```
┌──(kali㉿kali)-[~/playground]
└─$ rm fun-hard

┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 12
drwxr-xr-x 2 kali kali 4096 Apr 24 19:24 dir1
lrwxrwxrwx 1 kali kali    4 Apr 24 19:32 dir1-sym → dir1
drwxr-xr-x 2 kali kali 4096 Apr 24 19:28 dir2
-rw-r--r-- 3 kali kali 3217 Apr 18 17:49 fun
lrwxrwxrwx 1 kali kali    3 Apr 24 19:24 fun-sym → fun
```

Step 12: I removed the file and directory using the –i option

```
┌──(kali㉿kali)-[~/playground]
└─$ rm -i fun
rm: remove regular file 'fun'? yes

┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 8
drwxr-xr-x 2 kali kali 4096 Apr 24 19:24 dir1
lrwxrwxrwx 1 kali kali    4 Apr 24 19:32 dir1-sym → dir1
drwxr-xr-x 2 kali kali 4096 Apr 24 19:28 dir2
lrwxrwxrwx 1 kali kali    3 Apr 24 19:24 fun-sym → fun
```

Step 13: Here links that are left.

```
┌──(kali㉿kali)-[~/playground]
└─$ less fun-sym
fun-sym: No such file or directory

┌──(kali㉿kali)-[~/playground]
└─$ rm fun-sym dir1-sym

┌──(kali㉿kali)-[~/playground]
└─$ ls -l
total 8
drwxr-xr-x 2 kali kali 4096 Apr 24 19:24 dir1
drwxr-xr-x 2 kali kali 4096 Apr 24 19:28 dir2
```

Step 14: Playground has been removed from my system.

```
┌──(kali㉿kali)-[~/playground]
└─$ cd

┌──(kali㉿kali)-[~]
└─$ rm -r playground
```

Step 15: The following tells you the directory you are working in.

```
┌──(kali㉿kali)-[~]
└─$ whoami
kali

┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali
```

Step 16: I changed the current working directory.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ cd ..

┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali
```

Step 17: My current root directory is just kali

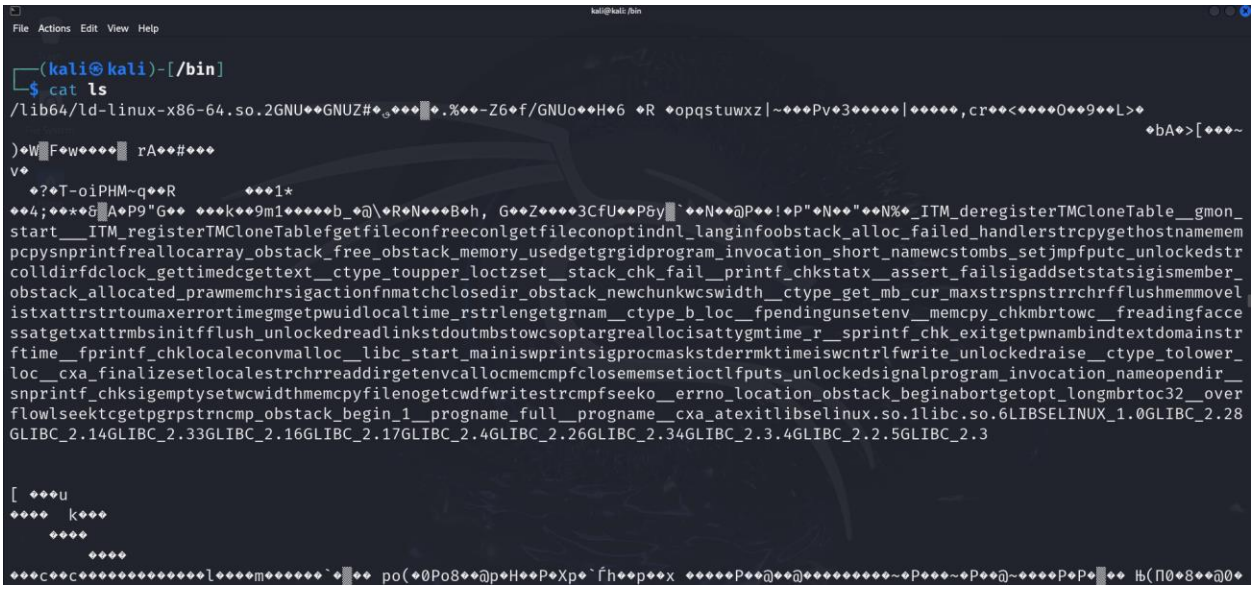
```
┌──(kali㉿kali)-[/home]
└─$ ls
kali
```

Step 18: This command helped me in accessing the binaries in my OS.

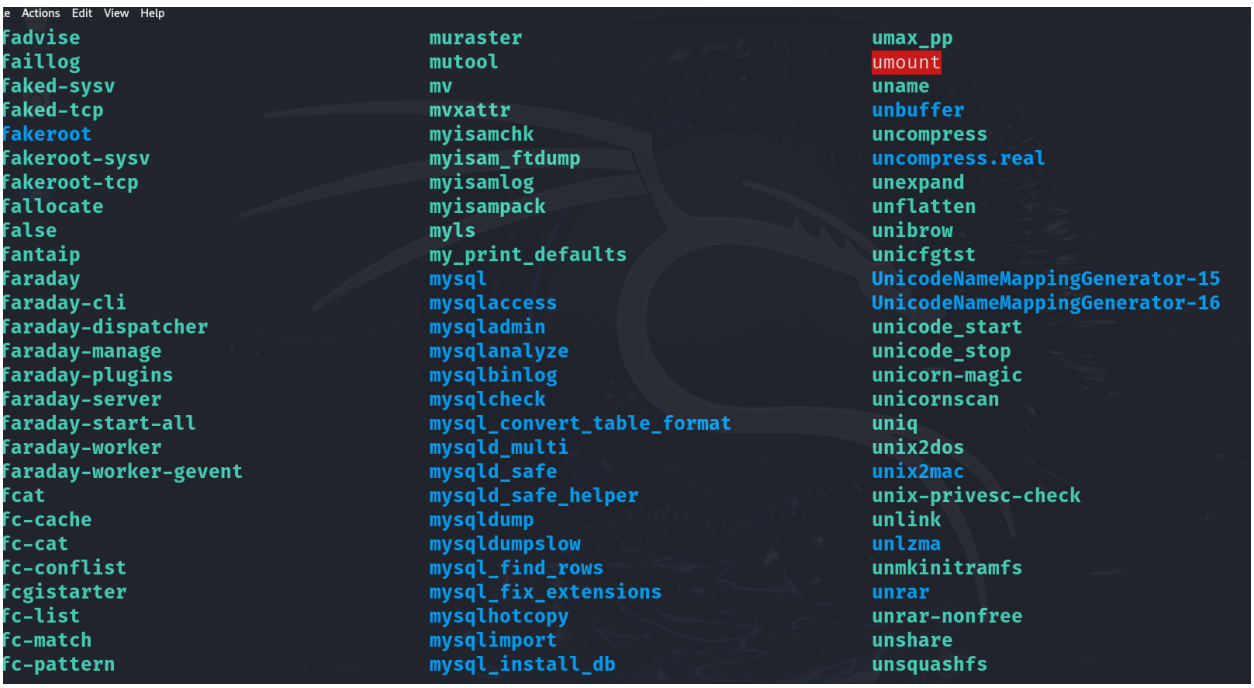
```
┌──(kali㉿kali)-[/home]
└─$ cd /bin

┌──(kali㉿kali)-[/bin]
└─$ ls
'['                                  invproj                          py3rsa-verify
 1password2john                      ionice                           py3versions
 2to3-2.7                            ios7tojohn                       pybabel
 7z                                  iostat                           pybabel-python3
 7z2john                             ip                               pyclean
 7za                                 ipcmk                            pycompile
 7zr                                 ipcrm                            pydoc
 aa-enabled                          ipcs                             pydoc2
 aa-exec                             iptables-xml                     pydoc2.7
 aa-features-abi                     ipwhois_cli.py                   pydoc3
 ab                                  ipwhois_utils_cli.py             pydoc3.11
 activate-global-python-argcomplete  ipython3                         pydoc3.12
 acyclic                             irb                              pyfiglet
 addpart                             irb3.1                           pygettext2
 addr2line                           ischroot                         pygettext2.7
 adxcsouf2john                       isdv4-serial-debugger            pygettext3
 aem2john                            isdv4-serial-inputattach         pygettext3.11
 afm2pl                              ispell-wrapper                   pygettext3.12
 afm2tfm                             istat                            pygmentex
 agentxtrap                          isympy                           pygmentize
```

Step 19: The following shows after using the cat ls command.



Step 20: Using the delete Ls command, here is the result.



Step 21: Here is the result of the return ls command.

```
fincore                          ncursesw6-config                 uptime
find                             ndisasm                          upx
find-debuginfo                   neato                            upx-ucl
findmnt                          neo2john                         urlscan
finger                           neqn                             us
firefox                          net                              usb-devices
firefox-esr                      netaddr                          usbhid-dump
fiwalk                           netcat                           usbreset
fixproc                          netkey-tool                      users
flac                             netkit-rcp                       users-admin
flask                            netkit-rlogin                    usnjls
flex                             netkit-rsh                       utmpdump
flex++                           netmask                          vacuumdb
flock                            netsed                           vacuumlo
fls                              net-server                       varlinkctl
fmt                              net-snmp-create-v3-user          VBoxClient
fmtutil                          netstat                          vbutil_firmware
fmtutil-sys                      networkctl                       vbutil_kernel
fmtutil-user                     newgrp                           vbutil_key
fold                             ngettext                         vbutil_keyblock
fontdump                         ngrep                            vdi2john
fonttosfnt                       nice                             vdir
fping                            nikto                            verify-uselistorder-14
fping6                           nisdomainname                    verify-uselistorder-15
```

Step 22: Here is the content.

```
┌──(kali㉿kali)-[/bin]
└─$ cat cat
@@@@◆       IPIP◆◆◆◆!◆!0◆0◆0◆p888◆ XXXDDS◆td888 P◆td◆◆◆LLQ◆tdR◆td0◆0◆0◆◆◆/lib64/ld-linux-x86-64.so.2GNU◆◆GNUy◆◆◆w*P◆◆}[o
%h\◆◆M◆,GNU?I◆ ?AF(◆◆e◆m◆Pv◆◆◆◆◆◆,cr◆bA◆9◆◆3◆◆◆jM<t ;◆ ◆"◆E◆A◆Z◆◆dS◆/h◆◆|◆_-◆◆◆◆(K8 c◆%◆u◆◆^5◆<!PG ◆◆n◆◆"◆◆!├!◆◆optindnl_la
nginfogetpagesizereallocarrayprogram_invocation_short_namefputc_unlockeddcgettext__stack_chk_fail__printf_chkfree__fpurge__cty
pe_get_mb_cur_maxstrrchrfflushmemmoveerrorstrlen__ctype_b_loc__fpendingread__freadingclearerr_unlockedmbsinitfflush_unlockedst
doutreallocposix_fadvise_exitbindtextdomain__fprintf_chkopenmalloc__libc_start_maincopy_file_rangeiswprintstderraligned_alloc_
_cxa_finalizesetlocalecallocstpcpymemcmpfclosememsetioctlfputs_unlockedprogram_invocation_namememcpyfilenofwritestrcmpunamefse
eko__errno_locationabortgetopt_longmbrtoc32lseekfstatstrncmp__progname_full__progname__cxa_atexitlibc.so.6GLIBC_2.3GLIBC_2.33G
LIBC_2.3.4GLIBC_2.14GLIBC_2.16GLIBC_2.27GLIBC_2.4GLIBC_2.26GLIBC_2.34GLIBC_2.2.5_ITM_deregisterTMCloneTable__gmon_start___ITM_
registerTMCloneTable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◆◆◆◆◆◆◆◆u i◆◆◆◆0028◆1@l◆`|◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆ ◆◆@ã◆◆◆◆◆◆◆◆◆"◆◆6◆◆8◆◆◆@◆◆◆◆◆◆ ◆◆◆◆ ◆◆◆◆◆ş%n8@◆◆A◆◆?◆◆F◆◆C◆◆E◆◆ ◆ ◆(0◆       8◆
@◆
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X◆`◆h◆p◆x◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆
```

```
File  Actions  Edit  View  Help
└─$ /sbin

┌──(kali㉿kali)-[/sbin]
└─$ ls
a2disconf                flash_unlock        mkfs.cramfs          select-default-wor
a2dismod                 flowtop             mkfs.exfat           sensors-detect
a2dissite                fmaptool            mkfs.ext2            serve_image
a2enconf                 fsck                mkfs.ext3            service
a2enmod                  fsck.cramfs         mkfs.ext4            setcap
a2ensite                 fsck.exfat          mkfs.fat             setvesablank
a2query                  fsck.ext2           mkfs.jffs2           sfdisk
aa-remove-unknown        fsck.ext3           mkfs.minix           sgdisk
aa-status                fsck.ext4           mkfs.msdos           shadowconfig
aa-teardown              fsck.fat            mkfs.ntfs            showmount
accessdb                 fsck.minix          mkfs.ubifs           shutdown
addgnupghome             fsck.msdos          mkfs.vfat            SIPdump
addgroup                 fsck.vfat           mkhomedir_helper     slattach
add-shell                fsfreeze            mkinitramfs          smartctl
adduser                  fstab-decode        mklost+found         smartd
agetty                   fstrim              mkntfs               smbd
airbase-ng               ftl_check           mkswap               sm-notify
aireplay-ng              ftl_format          mkvcalcproba         snmpd
airmon-ng                gdisk               ModemManager         split-logfile
airodump-ng              genccode            modinfo              sshd
airodump-ng-oui-update   gencmn              modprobe             sshmitm
airserv-ng               genl                mosquitto            sshow
airtun-ng                genmkvpwd           mount.cifs           sslh
```

Step 25: A new user has been created.

```
┌──(kali㉿kali)-[/sbin]
└─$ sudo adduser chris
info: Adding user `chris' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `chris' (1001) ...
info: Adding new user `chris' (1001) with group `chris (1001)' ...
info: Creating home directory `/home/chris' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N] y
New password:
Retype new password:
```

Step 26: I was able to access and see the contents.

```
┌──(kali㉿kali)-[~]
└─$ /usr

┌──(kali㉿kali)-[/usr]
└─$ ls
bin  games  i686-w64-mingw32  include  lib  lib32  lib64  libexec  local  sbin  share  src  x86_64-w64-mingw32  x86_64-w64-mingw32ucrt
```

Step 27: Here are the results.

```
┌──(kali㉿kali)-[/usr]
└─$ which ls /bin/ls
ls: aliased to ls --color=auto
/bin/ls

┌──(kali㉿kali)-[/usr]
└─$ which is cat /bin/cat
is not found
/usr/bin/cat
/bin/cat

┌──(kali㉿kali)-[/usr]
└─$ which is myls /bin/myls
is not found
/usr/bin/myls
/bin/myls
```

Step 28: This command helped me in displacing the contents of the library.

```
┌──(kali㉿kali)-[/usr]
└─$ /usr/lib

┌──(kali㉿kali)-[/usr/lib]
└─$ ls
7zip                        GraphicsMagick-1.3.42                libgvm-pg-server.so.23      policykit-1-gnome
apache2                     groff                                libgvm-pg-server.so.23.3.0  polkit-1
apparmor                    grub                                 libhotpatch.a               postgresql
apt                         grub-legacy                          libhotpatch.so              pppd
aspell                      gvfs                                 libhotpatchtest.so          pulse-16.1+dfsg1
bfd-plugins                 hashcat                              libopenvas_misc.so          python2.7
binfmt.d                    hashcat-utils                        libopenvas_misc.so.22       python3
blt2.5                      hdparm                               libopenvas_misc.so.22.7.9   python3.11
cgi-bin                     ifupdown                             libopenvas_nasl.so          python3.12
chromium                    init                                 libopenvas_nasl.so.22       qt5
clang                       ipsec                                libopenvas_nasl.so.22.7.9   qt6
cmake                       ispell                               libsupp.a                   rpm
cnf-update-db               java-wrappers                        linux                       ruby
command-not-found           john                                 linux-boot-probes           runit-helper
compat-ld                   jvm                                  llvm-14                     sasl2
console-setup               kali_tweaks                          llvm-15                     security
cpp                         kernel                               llvm-16                     sftp-server
cryptsetup                  klibc                                locale                      speech-dispatcher-modules
cryptsetup-nuke-password    klibc-D67x9cQZJoe6j863VWhTpYmFGDU.so lsb                         ssl
cups                        ld-linux.so.2                        man-db                      sysctl.d
dbus-1.0                    libarmadillo.so.12                   mime                        systemd
debug                       libarmadillo.so.12.6.7               modprobe.d                  sysusers.d
```

Step 29: This command should help me see the games I games, but I do not have any games on my Linux.

```
┌──(kali㉿kali)-[/usr/lib]
└─$ /usr/games

┌──(kali㉿kali)-[/usr/games]
└─$ ls
```

Step 30: In this step, Im able to see the booting.

```
┌──(kali㉿kali)-[/boot]
└─$ /boot

┌──(kali㉿kali)-[/boot]
└─$ ls
config-6.5.0-kali3-amd64  grub                      initrd.img-6.6.9-amd64        System.map-6.6.9-amd64     vmlinuz-6.6.9-amd64
config-6.6.15-amd64       initrd.img-6.5.0-kali3-amd64 System.map-6.5.0-kali3-amd64 vmlinuz-6.5.0-kali3-amd64
config-6.6.9-amd64        initrd.img-6.6.15-amd64     System.map-6.6.15-amd64       vmlinuz-6.6.15-amd64
```

Step 31: On this step, Im able to view the content under this library

```
┌──(kali㉿kali)-[/boot]
└─$ /tmp

┌──(kali㉿kali)-[/tmp]
└─$ ls
hsperfdata_root
ssh-AUX3WH1qtSDW
systemd-private-0cdbe80e64fc4b7a897fab00d4d84e6c-colord.service-sndxct
systemd-private-0cdbe80e64fc4b7a897fab00d4d84e6c-haveged.service-WICqHu
systemd-private-0cdbe80e64fc4b7a897fab00d4d84e6c-ModemManager.service-S1pAZg
systemd-private-0cdbe80e64fc4b7a897fab00d4d84e6c-polkit.service-gRjMYE
systemd-private-0cdbe80e64fc4b7a897fab00d4d84e6c-systemd-logind.service-PLtowW
systemd-private-0cdbe80e64fc4b7a897fab00d4d84e6c-upower.service-qfh78w
tmpaddon
```

Step 32: This command takes me back to my home and with ls I'm able to see its content.

```
┌──(kali㉿kali)-[/tmp]
└─$ /home

┌──(kali㉿kali)-[/home]
└─$ ls
chris  kali
```

Step 33: I became root and was allowed to view the contents with the ls command.

```
┌──(kali㉿kali)-[/]
└─$ sudo ls root
[sudo] password for kali:

┌──(kali㉿kali)-[/]
└─$ ls
bin   dev  home        initrd.img.old  lib32  lost+found  mnt  proc  run   srv       sys  usr  vmlinuz
boot  etc  initrd.img  lib             lib64  media       opt  root  sbin  swapfile  tmp  var  vmlinuz.old
```

Step 34: Here Im allowed to view connections to my system,

```
┌──(kali㉿kali)-[/]
└─$ /dev

┌──(kali㉿kali)-[/dev]
└─$ ls
autofs          dri        loop0         null    sg0      tty11  tty23  tty35  tty47  tty59  uhid         vcs7   vcsu2
block           fb0        loop1         nvram   sg1      tty12  tty24  tty36  tty48  tty6   uinput       vcs8   vcsu3
bsg             fd         loop2         port    shm      tty13  tty25  tty37  tty49  tty60  urandom      vcsa   vcsu4
btrfs-control   full       loop3         ppp     snapshot tty14  tty26  tty38  tty5   tty61  userfaultfd  vcsa1  vcsu5
bus             fuse       loop4         psaux   snd      tty15  tty27  tty39  tty50  tty62  vboxguest    vcsa2  vcsu6
cdrom           hidraw0    loop5         ptmx    sr0      tty16  tty28  tty4   tty51  tty63  vboxuser     vcsa3  vcsu7
char            hpet       loop6         pts     stderr   tty17  tty29  tty40  tty52  tty7   vcs          vcsa4  vcsu8
console         hugepages  loop7         random  stdin    tty18  tty3   tty41  tty53  tty8   vcs1         vcsa5  vfio
core            hwrng      loop-control  rfkill  stdout   tty19  tty30  tty42  tty54  tty9   vcs2         vcsa6  vga_arbiter
cpu             initctl    mapper        rtc     tty      tty2   tty31  tty43  tty55  ttyS0  vcs3         vcsa7  vhci
cpu_dma_latency input      mem           rtc0    tty0     tty20  tty32  tty44  tty56  ttyS1  vcs4         vcsa8  vhost-net
cuse            kmsg       mqueue        sda     tty1     tty21  tty33  tty45  tty57  ttyS2  vcs5         vcsu   vhost-vsock
disk            log        net           sda1    tty10    tty22  tty34  tty46  tty58  ttyS3  vcs6         vcsu1  zero
```

Step 35: For the cat sda command, this pops up

```
hY\◆Z\H\\◆]\(_\◆`b\`c\◆d\@f\◆g\ i\◆j\l\◆m\◆n\hp\◆q\Hs\◆t\(v\◆wy\`z\◆{\@}\◆~\ ◆\◆◆\◆\◆◆\◆◆\h◆\◆◆\H◆\◆◆\(◆\◆◆◆\`◆\◆◆\@◆\ü\ ◆\Ş\◆◆\◆◆◆\◆◆\
8◆\ ◆\p◆\p◆\`◆\(◆\◆◆\`◆\`◆\8◆\◆\◆◆\◆◆\◆◆\X◆\ ◆\h◆\◆◆\ ◆\◆◆\h◆\◆◆\`◆\◆◆\h◆\◆◆\P◆\@◆\◆◆\◆◆\◆◆\X◆\x◆\◆◆◆\x◆\◆◆\(◆\◆◆\◆◆\◆◆\▒]8](]◆]◆]X
](
]◆]◆]x]0▒]▒]◆]◆]P!] #]P%]◆&]◆+] .]P/]◆3]`5]◆6]◆79]◆:]◆;]X>]◆?]◆@]◆A]▒C]◆D]◆F]8I]◆J]◆M]◆P]◆R](T]◆U](W]XZ]◆[]◆^]Xa]0c]◆d]◆g]pi]◆j]◆l](n]x
o]◆q]◆ru]◆x]◆y]◆{]p}] ]h◆]◆]◆]8◆]◆◆](◆]p◆]`◆]◆◆]◆◆]▒◆]h◆]◆◆]8◆]◆◆]p◆]◆◆]H◆]p◆]◆◆]◆]◆◆]8◆]◆◆]▒◆]◆◆]0◆]◆◆]◆◆]X◆]Ÿ]8◆]◆◆]P◆]◆◆]@◆]◆◆]X◆]H◆
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^◆^◆^◆^◆^◆^◆▒^◆▒^◆^◆ "^@/^◆◇^◆A^8C^◆e^◆g^◆
i^◆k^◆m^(p^Pr^◆~^◆◆^Ã^8◆^X◆^x◆^◆◆^(◆^H◆^◆^Ÿ^◆◆^8◆^Γ^◆◆^^◆◆^ţ^◆◆^X◆^x◆^◆◆^`◆^◆◆^8◆◆^◆◆^p◆^h◆^◆◆^◆^◆◆^`◆^◆◆^◆◆^◆◆^P◆^◆◆^◆◆^◆◆^◆◆^◆^0◆^`◆^
◆◆^◆◆^◆^◆◆^◆◆^◆^◆◆^H◆^x◆^◆◆◆^◆◆^H◆^◆◆^X◆^0◆^◆◆^◆◆^◆◆◆^H◆^H◆^▒_◆_X_x_◆
_p
_◆_0_◆(_◆_◆_p!_◆#_P%_◆'_8)_+_◆,_8._◆/_`1_▒3_◆4_@6_9_◆;_`=?_◆@_0C_◆D_HF_H_◆I_(KM_◆N_◆P_pR_▒T_◆U_xW_ Y_◆Z_◆\_◆^_0`_◆a_◆c_`e_◆g_◆i_◆k_◆m_o
_◆q_Ps_0u_◆v_(x_▒z_◆|_◆_H◆_Ъ_(◆_◆◆◆_◆◆_◆◆_p◆_Ö_P◆_◆◆_0◆_◆◆_◆_h◆_◆◆_H◆_Й_(◆_◆◆◆_◆◆_◆◆_p◆_Ø_P◆_◆◆_0◆_◆◆_◆_h◆_◆◆_H◆_a_(◆_◆◆◆_◆◆_◆◆_p◆_Ä_P◆
_◆◆_0◆_◆◆_◆_h◆_◆◆_H◆_◆◆_(◆_◆◆◆_◆◆_◆◆_p◆_◆◆_P◆_◆◆_0◆_◆◆_◆_h◆_◆◆_H◆_◆◆_(◆_◆◆◆_◆◆_◆◆_P◆_◆◆_0◆_◆◆_◆_h◆_◆◆_H◆_◆◆_(◆_◆◆◆_◆◆_◆◆_p◆_◆◆_P`
◆`0`◆```◆ `H
            `◆
              `(`◆`◆`◆`p`◆`P▒`◆`0◆``h`◆ `H"`◆#`(%`◆&(`◆)`◆*`p,`◆-`P/`◆0`02`◆3`5`h6`◆7`H9`◆:`(<`◆=?`◆@`◆A`pC`◆D`PF`◆G`0I`◆J`L`hM`◆N`HP`◆
Q`(S`◆TV`◆W`◆X`pZ`◆[`P]`◆^`0``◆a`c`hd`◆e`Hg`◆h`(j`◆km`◆n`◆o`pq`◆r`Pt`◆u`0w`◆x`z`h{`◆|`H~`◆`(◆`◆◆◆`◆◆`◆◆`p◆`ì`P◆`◆◆`0◆`◆◆`◆`h◆`◆◆`H◆`Ж`(
◆`◆◆◆`◆◆`◆◆`p◆`Π`P◆`◆◆`0◆`◆◆`◆`h◆`◆◆`H◆`Э`(◆`◆◆◆`◆◆`◆◆`p◆`j`P◆`◆◆`0◆`◆◆`◆`h◆`◆◆`H◆`◆◆`(◆`◆◆◆`◆◆`◆◆`p◆`◆◆`P◆`◆◆`0◆`◆◆`◆◆`◆◆`◆◆`◆◆`p◆`x◆`◆axa◆a◆a`  a`
a◆a◆a◆a◆a◆a`a(◆a◆a◆ aH#a◆%a◆&a◆(a◆*a◆+a◆-a◆0a◆2aP4a◆6aX8a :a8<ap>a8@a◆Ba◆CaXEaHa◆IaKa◆La◆Ma◆Na`Pa◆QTa◆Va◆Xap[a◆]a▒_a◆aa◆ca◆dafa◆ha◆ja0m
aoa◆qahsa`ua◆va@yapza▒|aH}ax~a◆aX◆a◆◆a◆◆a◆◆ah◆a◆◆a◆◆a0◆a▒◆a◆◆a_a▒◆aЪa◆◆a▒◆ax◆a◆◆ax◆a◆◆a_a(◆a(◆aжaH◆a◆◆a◆◆a◆a◆◆ap◆aH◆a◆◆a◆◆ap◆aH◆a ◆a◆◆a
◆◆a◆◆a◆◆aX◆a0◆◆a◆◆a◆◆a◆◆ax◆aP◆a(◆a◆a◆a ◆a◆◆a(◆a◆a◆a ◆a0◆a@◆a◆◆a◆◆a`◆ap◆a◆◆◆aP◆a◆◆a◆◆a8◆a◆a◆Mb◆NbhOb@PbPQb`RbpSb◆Tb Ub Vb0Wb@XbPYb`Zb
p[b◆\b◆]b◆^bx_b◆`b`ab8bbcb◆cb◆db◆eb◆fb◆gbphbHib jb kb◆kbHmb◆nbpobppbpqbprbpsbptbHub vb◆vb◆wb◆xb◆ybXzb0{▒b◆|b◆}b◆~bhb@◆b▒◆b◆◆bÂb◆◆bx◆b▒◆
bp◆b◆b⅄b◆◆b@◆b▒◆b◆◆bÔb◆◆bx◆bP◆b(◆b◆bb◆◆b◆◆b`◆b8◆b◆b◆◆b◆◆b◆◆bp◆bH◆b ◆b◆◆bOb◆◆b◆◆bX◆b0◆◆b◆◆b◆◆bh◆b@◆b▒◆b◆◆bppcHqc rc`sc8tcuc◆uc◆vc◆wcp
xcHyc zc◆zc◆{c◆|c◆}cX~c0◆c◆◆c◆◆c◆◆ch◆c@◆c▒◆c◆◆cĒc◆◆cx◆cP◆c(◆c◆c$c◆b(PT ◆Uh◆U@◆U◆◆U ◆Ux◆U◆UX◆U◆◆U8◆U◆◆U▒◆U◆◆U◆◆U◆◆U`◆U◆◆U@◆U◆◆U ◆Ux◆U
◆UX◆U◆V8V◆V▒V◆V◆V◆  V◆
V`
V@V◆V VxVVXV◆V8V◆▒V▒V◆V◆V◆ V◆!V`#V◆$V@&V◆'V )Vx*V,VX-V◆.V80V◆1V▒3V◆4V◆5V◆7V◆8V`:V◆;V@=V◆>V @VxAVCVXDV◆EV8GV◆HV▒JV◆KV◆LV◆NV◆OV`QV◆RV@TV◆
```

Step 36: In this step, Im able to view files in my Linux

```
┌──(kali㉿kali)-[~]
└─$ /etc

┌──(kali㉿kali)-[/etc]
└─$ ls
adduser.conf                  debtags                hosts.allow          login.defs           openfortivpn         request-key.d        supercat
alsa                          default                hosts.deny           logrotate.conf       openni2              resolv.conf          sv
alternatives                  deluser.conf           idmapd.conf          logrotate.d          opensc               responder            sysctl.conf
apache2                       dhcp                   ifplugd              macchanger           openvas              rmt                  sysctl.d
apparmor                      dictionaries-common    ImageMagick-6        machine-id           openvpn              rpc                  sysstat
apparmor.d                    dns2tcpd.conf          inetsim              magic                opt                  runit                systemd
apt                           doc-base               init.d               magic.mime           os-release           samba                terminfo
arp-scan                      dpkg                   initramfs-tools      mailcap              pam.conf             sane.d               texmf
avahi                         e2scrub.conf           inputrc              mailcap.order        pam.d                scalpel              theHarvester
bash.bashrc                   eac                    insserv.conf.d       manpath.config       papersize            screenrc             tightvncserver.conf
bash_completion               emacs                  ipp-usb              matplotlibrc         passwd               sddm.conf.d          timezone
bash_completion.d             environment            ipsec.conf           mime.types           passwd-              searchsploit_rc      timidity
bindresvport.blacklist        environment.d          ipsec.d              minicom              perl                 security             tmpfiles.d
binfmt.d                      ethertypes             ipsec.secrets        miredo               php                  selinux              ts.conf
bluetooth                     ettercap               issue                miredo.conf          plymouth             sensors3.conf        ucf.conf
ca-certificates               firebird               issue.net            mke2fs.conf          polkit-1             sensors.d            udev
ca-certificates.conf          firefox-esr            java-17-openjdk      ModemManager         postgresql           services             udisks2
ca-certificates.conf.dpkg-old fonts                  java-21-openjdk      modprobe.d           postgresql-common    sgml                 ufw
chatscripts                   freetds                john                 modules              powershell-empire    shadow               unicornscan
chromium                      fstab                  kali-menu            modules-load.d       ppp                  shadow-              updatedb.conf
chromium.d                    fuse.conf              kernel               mosquitto            profile              shells               update-motd.d
cifs-utils                    gai.conf               keyutils             motd                 profile.d            skel                 UPower
cloud                         geoclue                kismet               mtab                 protocols            smartd.conf          usb_modeswitch.conf
colord                        ghostscript            ldap                 mysql                proxychains4.conf    smartmontools        usb_modeswitch.d
console-setup                 glvnd                  ld.so.cache          nanorc               pulse                smi.conf             vconsole.conf
cracklib                      gnome-system-tools     ld.so.conf           netconfig            python2.7            snmp                 vdpau_wrapper.cfg
credstore                     gophish                ld.so.conf.d         netsniff-ng          python3              speech-dispatcher    vim
credstore.encrypted           gprofng.rc             legion.conf          network              python3.11           sqlmap               vpnc
```

Questions:

1. Command line programs often require fewer system resources and can be faster to use for experienced users and command line programs are easily scriptable, allowing for automation of tasks. But the disadvantages are that Using command line programs can have a steep learning curve for beginners and Some commands can be complex and require memorization of options and arguments.

2. In Linux, a wildcard is a character that represents one or more other characters. Wildcards are used with commands like ls, cp, mv, and rm to specify groups of files or directories. Common wildcards include * (matches any number of characters), ? (matches a single character), and [] (matches a range of characters). For example, *.txt would match all files with a .txt extension in a directory.

3. For hard links, they cannot span across different filesystems or partitions. Removing a hard link does not remove the original file's data unless it was the last hard link. For symbolic links they point to the path of another file or directory. Can span across different filesystems or partitions.

4. /bin, /usr/bin, /sbin,/etc, /usr/sbin

5. /bin and /usr/bin: Both directories contain executable binaries, but /bin contains essential binaries required for system boot and repair, while /usr/bin contains binaries for regular user applications and utilities.

/sbin and /usr/sbin: Similarly, /sbin contains system administration binaries required for boot and repair, while /usr/sbin contains non-essential system administration binaries.

Conclusion: In this lab, I used the given commands of the pdf to have a more hands on practice on how to manipulate files and directories using top level directory commands, removing and moving files, deleting directories. Overall, after this lab, I became more proficient and well versed in the commands and will know which one to use in future jobs and interships.

References:

Linux files - javatpoint. www.javatpoint.com. (n.d.). https://www.javatpoint.com/linux-files

GeeksforGeeks. (2023, June 8). *Linux directory structure*. https://www.geeksforgeeks.org/linux-directory-structure/